

Here's a **10-mark summarized answer** on **Hebbian Learning** based on the contents of your PDF:

Hebbian Learning – 10 Marks Summary

Definition:

Hebbian Learning is a **biologically inspired learning rule** proposed by **Donald Hebb (1949)**. It states:

“When a neuron A repeatedly helps fire neuron B, the connection between them strengthens.”

Mathematically, the change in weight is:

$$\Delta w_i(t) \propto x_i(t) \cdot y(t)$$

Where:

- $x_i(t)$ = input
- $y(t)$ = output
- $\Delta w_i(t)$ = change in weight

Working Principle:

- **Neurons** strengthen their connection when activated together.
- In vector form:

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$$\Delta \vec{w} = \eta V \vec{\xi}$$

Where η is learning rate, V is output, and $\vec{\xi}$ is input vector.

Key Properties:

1. **Unsupervised learning**— no need for labeled output.
 2. **Locality**— only uses data from connected neurons.
 3. **Biologically plausible**— aligns with observed brain behavior.
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Limitation:

- **Weight explosion:** Weights keep growing indefinitely unless controlled.
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Stabilization Techniques:

1. **Weight Normalization**— Normalize the weight vector after each update.
2. **Oja's Rule**— Adds a decay term to prevent weight growth:

2. **Oja's Rule** – Adds a decay term to prevent weight growth:

$$\Delta \vec{w} = \eta V (\vec{\xi} - V \vec{w})$$

3.

Sejnowski's Covariance Rule— Adjusts learning by subtracting the means.

Relation to PCA:

- Hebbian learning aligns the weight vector to the **direction of maximum variance** in the data.
 - This is the **first principal component** in **Principal Component Analysis (PCA)**.
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Applications:

- **Neural modeling**
 - **Feature extraction**
 - **Principal component learning**
 - **Visual system simulations (e.g., Linsker's model)**
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Let me know if you'd like diagrams or a handwritten-style version for class notes!

The Hebbian Synapse

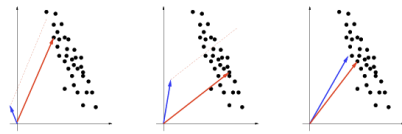
A biological system which exhibits Hebbian learning.



$$\Delta w_1(t) \propto x(t) y(t)$$

Hebbian Learning, Non-zero-mean Data

Recall that $\Delta \bar{w} = \eta V \bar{\xi}$ and that $V = \bar{w}^T \bar{\xi}$.



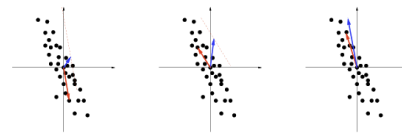
But note that if the initial $\bar{w}$ points away from the cluster, the dot product with $\bar{\xi}$ will be negative, and $\bar{w}$ will ultimately point in exactly the opposite direction.

Interpretation awkward ...

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Plain Hebbian Learning, Zero-mean Data

Recall that $\Delta \bar{w} = \eta V \bar{\xi}$ and that $V = \bar{w}^T \bar{\xi}$.



In this case, the weight vector will ultimately align itself with the *direction of greatest variance* in the data.

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Hebbian vs Competitive Learning

Plain Competitive Learning	Plain Hebbian Learning
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Similarities

Unsupervised training methods (no teacher, no error signal).
Frequently used with online learning.
Strong connection to biological systems.
Linear units: $V = \bar{w}^T \bar{\xi}$

Differences

Exactly one output must be active.	No constraint imposed by neighboring units.
Only winner's weights updated at every epoch.	All weights updated at every epoch.
$\Delta \bar{w} = \eta \bar{\xi}$	$\Delta \bar{w} = \eta V \bar{\xi}$